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Batch- 2024-2028

Subject- Wireshark assignment using the provided PCAP
(network capture) file

[illegible]

Ans-5

E2. Analyze the network file and find the number of ARP messages.

The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets, with ARP messages highlighted in orange. The middle pane shows the details of the selected ARP message (No. 1467), including the Ethernet II header and the ARP request payload. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1414	63.376310	HewlettPacka_9e...	Broadcast	ARP	60	Who has 10.103.51.22? Tell 10.103.50.221
1419	63.582677	BrotherIndus_93...	Broadcast	ARP	60	Who has 10.103.50.37? Tell 10.103.200.10
1430	63.834024	Intel_6c:26:0a...	Broadcast	ARP	60	Who has 10.103.200.5? Tell 10.103.0.18
1444	64.007075	HewlettPacka_a6...	Broadcast	ARP	60	Who has 10.103.50.88? Tell 10.103.50.249
1450	64.235454	HewlettPacka_9b...	Broadcast	ARP	60	Who has 10.103.51.22? Tell 10.103.50.8
1460	64.441683	Intel_6c:26:0a...	Broadcast	ARP	60	Who has 10.103.200.5? Tell 10.103.0.18
1466	64.630428	HewlettPacka_a6...	Broadcast	ARP	60	Who has 10.103.50.88? Tell 10.103.50.249
1467	64.758799	HewlettPacka_9b...	Broadcast	ARP	60	Who has 10.103.51.22? Tell 10.103.50.8
1482	64.966054	HewlettPacka_78...	Broadcast	ARP	42	Who has 10.103.0.1? Tell 10.103.51.159
1485	64.966238	CheckpointSo_6c...	HewlettPacka_78...	ARP	60	10.103.0.1 is at 00:1c:7f:6c:96:3f
1508	65.027773	HewlettPacka_78...	Broadcast	ARP	42	Who has 10.103.51.159? (ARP Probe)
1519	65.218885	HewlettPacka_78...	Broadcast	ARP	42	Who has 10.103.0.1? Tell 10.103.51.159
1520	65.219018	CheckpointSo_6c...	HewlettPacka_78...	ARP	60	10.103.0.1 is at 00:1c:7f:6c:96:3f
1524	65.440081	Intel_6c:26:0a...	Broadcast	ARP	60	Who has 10.103.200.5? Tell 10.103.0.18
1533	65.630400	HewlettPacka_a6...	Broadcast	ARP	60	Who has 10.103.50.88? Tell 10.103.50.249
1536	65.763554	HewlettPacka_9b...	Broadcast	ARP	60	Who has 10.103.51.22? Tell 10.103.50.8

Frame 1467: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0, Ethernet II, Src: HewlettPacka_9b:73:89 (a0:14:81:c9:73:89), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

Answer: 680

E3. Analyze the network file and find the IP address that accessed 'baidu.com'.

traffic: pcapng									
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display name contains 'baidu'									
No.	Time	Source	Destination	Protocol	Length	Info			
20754	305.1187359	10.183.0.20	10.183.51.159	DNS	142	Standard query response 008726 A www.baidu.com			
20755	305.1187359	10.183.0.20	10.183.51.159	DNS	142	Standard query response 008726 A www.baidu.com			
20813	304.1311842	10.183.51.159	10.183.0.20	DNS	73	Standard query 001440 A www.baidu.com			
20825	304.2031192	10.183.0.20	10.183.51.159	DNS	142	Standard query response 001440 A www.baidu.com			
20843	304.2037766	10.183.0.20	10.183.51.159	DNS	73	Standard query 002746 A www.baidu.com			
20844	304.2037766	10.183.0.20	10.183.51.159	DNS	73	Standard query 002746 A www.baidu.com			
20838	304.207966	10.183.51.159	10.183.0.20	DNS	73	Standard query 004523 A www.baidu.com			
20829	304.209291	10.183.0.20	10.183.51.159	DNS	142	Standard query response 004523 A www.baidu.com			
20894	304.880625	10.183.51.159	10.183.0.20	DNS	73	Standard query 001643 A www.baidu.com			
20895	304.880625	10.183.51.159	10.183.0.20	DNS	73	Standard query 001643 A www.baidu.com			
20896	304.880625	10.183.51.159	10.183.0.20	DNS	73	Standard query 001643 A www.baidu.com			
20897	304.880688	10.183.51.159	10.183.0.20	DNS	82	Standard query 007318 A www.baidu.com			
20946	305.249358	10.183.0.20	10.183.51.159	DNS	116	Standard query response 00936c A www.baidu.com			
20959	305.323084	10.183.0.20	10.183.51.159	DNS	132	Standard query response 004157 A www.baidu.com			
20970	305.324575	10.183.51.159	10.183.0.20	DNS	76	Standard query 000973 A www.baidu.com			
20971	305.324575	10.183.51.159	10.183.0.20	DNS	131	Standard query response 000973 A www.baidu.com			
20972	305.331613	10.183.51.159	10.183.0.20	DNS	74	Standard query 007276 A www.baidu.com			
20977	305.395249	10.183.0.20	10.183.51.159	DNS	259	Standard query response 007276 A www.baidu.com			
20978	305.405844	10.183.51.159	10.183.0.20	DNS	74	Standard query 00648c A www.baidu.com			
20979	305.405844	10.183.51.159	10.183.0.20	DNS	122	Standard query response 00648c A www.baidu.com			
> Frame 20741: Packet: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface DeviceNPF_{...}									
> Ethernet II, Src: VMwareNet76 (08:00:2b:64:10:08), Dst: VMwareNet60 (08:00:2b:64:10:08)									
> Internet Protocol Version 4, Src: 10.183.51.159, Dst: 10.183.0.20									
> User Datagram Protocol, Src Port: 62733, Dst Port: 53									
> Domain Name System (query)									

Answer: 10.103.51.159

[illegible]

E4. Analyze the network file and find the number of packets the IP address 10.103.0.20 sent.

[illegible]

Answer: 496 packets

E5. Analyze the network file and find the number of UDP packets.

[illegible]

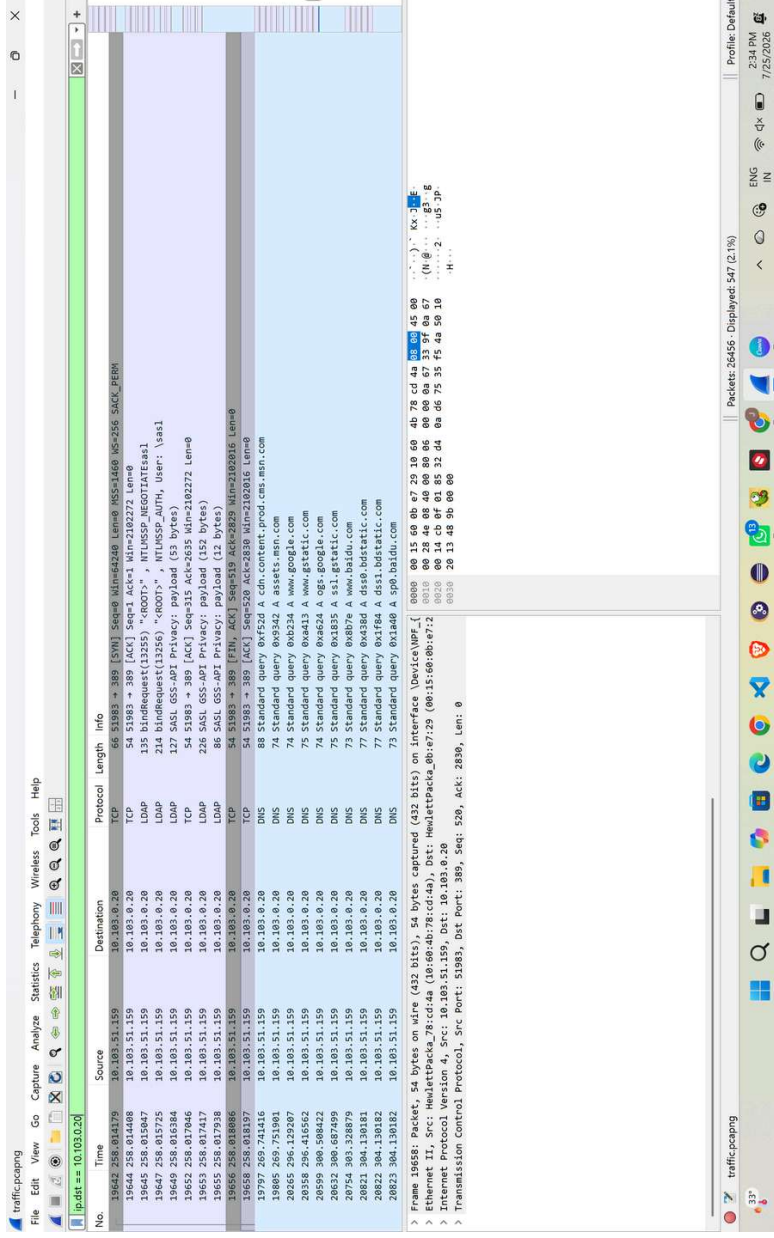
Answer: 9102 UDP packets

E6. Analyze the network file and find the number of SMB packets

[illegible]

Ans: 518

E7. Analyze the network file and find the number of packets sent to the source
IP address 10.103.0.20.



Ans:547

E8. Examine the network file and isolate only the IPv4 traffic; determine the count of packets shown.

traffic-capturing

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IP

No.	Time	Source	Destination	Protocol	Length	Info
15649	258.016384	10.103.51.159	10.103.0.20	LDAP	127	SASL GSS-API Privacy: payload (53 bytes)
15650	258.016936	10.103.0.20	10.103.51.159	TCP	1514	889 → 51983 [ACK] Seq=255 Ach=315 Win=65221 Len=1460 [TCP PDU reassembled in 19651]
15651	258.017025	10.103.0.20	10.103.51.159	LDAP	974	SASL GSS-API Privacy: payload (1360 bytes)
15652	258.017046	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=315 Ach=2835 Win=2102272 Len=0
15653	258.017417	10.103.51.159	10.103.0.20	LDAP	226	SASL GSS-API Privacy: payload (152 bytes)
15654	258.017697	10.103.0.20	10.103.51.159	LDAP	248	SASL GSS-API Privacy: payload (174 bytes)
15655	258.017938	10.103.51.159	10.103.0.20	LDAP	86	SASL GSS-API Privacy: payload (12 bytes)
15656	258.018049	10.103.0.20	10.103.51.159	TCP	60	389 → 51983 [FIN, ACK] Seq=2835 Ach=520 Win=2102272 Len=0
15657	258.018164	10.103.0.20	10.103.51.159	TCP	60	389 → 51983 [FIN, ACK] Seq=2835 Ach=520 Win=2102272 Len=0
15658	258.018197	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=520 Ach=2830 Win=2102016 Len=0
15659	258.018282	10.103.0.20	10.103.51.159	TCP	60	389 → 51983 [ACK] Seq=2830 Ach=520 Win=65017 Len=0
15660	258.076292	10.103.51.159	12.0.1.28	TELNET	55	1 byte data
15662	258.165454	10.103.51.159	213.155.156.164	TCP	55	1 byte data
15664	258.188268	10.103.51.159	12.0.1.28	TELNET	55	1 byte data
15664	258.320822	12.0.1.28	10.103.51.159	TCP	60	23 → 58616 [ACK] Seq=1956 Ach=114 Win=65780 Len=0
15665	258.372222	10.103.51.159	12.0.1.28	TELNET	55	1 byte data
15666	258.416286	10.103.51.159	213.155.156.164	TCP	60	23 → 58616 [ACK] Seq=1956 Ach=115 Win=65780 Len=0
15667	258.426690	12.0.1.28	10.103.51.159	TCP	60	23 → 58616 [ACK] Seq=1956 Ach=116 Win=65780 Len=0
15668	258.610663	12.0.1.28	10.103.51.159	TCP	60	23 → 58616 [ACK] Seq=1956 Ach=116 Win=65780 Len=0
15669	258.676575	10.103.51.159	12.0.1.28	TELNET	55	1 byte data

> Frame 19658: Packet, 54 bytes on wire (432 bits), 54 bytes captured (0 on interface) on interface 0: [ethernet II, Src: Hewlett-Packard_78:cd:4a, Dst: Hewlett-Packard_00:15:60:0b:e7:2] [ethernet II, Src: 10.103.51.159, Dst: 10.103.0.20]

> Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20

> Transmission Control Protocol, Src Port: 389, Seq: 520, Ack: 2830, Len: 0

Internet Protocol Version 4: Protocol

Packets: 26456 - Displayed: 24906 (94.1%)

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Answer: 24906

source IP address.

Answer: Source | P:10.103.50.202

E10. Analyze the network file and find the MAC address of the IP 151.139.128.14.

The image shows a Wireshark network traffic capture. The top pane displays a list of 20 packets. The bottom pane shows the details of the selected packet (No. 19917), which is a TCP Reset (RST) from 151.139.128.14 to 10.103.51.159. The packet length is 60 bytes. The details pane shows the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header. The IP address 151.139.128.14 is highlighted in the packet list and the details pane.

No.	Time	Source	Destination	Protocol	Length	Info
19894	228.215524	10.103.51.159	151.139.128.14	TCP	60	64879 → 80 [SYN] Seq=0 Min=64208 Len=0 MSS=1460 WS=255 SACK_PERM=1
19895	228.215524	10.103.51.159	151.139.128.14	TCP	60	64879 → 80 [ACK] Seq=1 Min=64208 Len=0 MSS=1460 SACK_PERM=1 WS=128
19896	228.262845	10.103.51.159	151.139.128.14	HTTP	290	GET /HTTP/1.1 HTTP/1.1 200 OK (text/html) Content-Type: text/html; charset=UTF-8
19897	228.262845	10.103.51.159	151.139.128.14	HTTP	290	GET /HTTP/1.1 HTTP/1.1 200 OK (text/html) Content-Type: text/html; charset=UTF-8
19898	228.320882	151.139.128.14	10.103.51.159	TCP	60	80 → 64879 [ACK] Seq=1 Ack=237 Min=64128 Len=0
19899	228.321598	151.139.128.14	10.103.51.159	TCP	781	Response
19900	228.321598	151.139.128.14	10.103.51.159	OCSP	54	64879 → 80 [ACK] Seq=237 Ack=1173 Min=261376 Len=0
19901	228.321598	151.139.128.14	10.103.51.159	TCP	292	GET /HTTP/1.1 HTTP/1.1 200 OK (text/html) Content-Type: text/html; charset=UTF-8
19902	228.321598	151.139.128.14	10.103.51.159	HTTP	60	80 → 64879 [ACK] Seq=1173 Ack=475 Min=64128 Len=0
19903	228.321598	151.139.128.14	10.103.51.159	TCP	60	80 → 64879 [PSH, ACK] Seq=1173 Ack=475 Min=64128 Len=45 [TCP PDU reassembled in 19917]
19904	228.321598	151.139.128.14	10.103.51.159	TCP	563	Response
19905	228.321598	151.139.128.14	10.103.51.159	OCSP	54	64879 → 80 [ACK] Seq=475 Ack=2137 Min=262556 Len=0
19906	228.321598	151.139.128.14	10.103.51.159	TCP	54	64879 → 80 [FIN, ACK] Seq=475 Ack=2138 Min=262556 Len=0
19907	228.321598	151.139.128.14	10.103.51.159	TCP	60	80 → 64879 [ACK] Seq=2138 Ack=476 Min=64128 Len=0

Packet 19917 details:

- Ethernet II, Src: Hewlett-Packard (10:00:00:00:00:00), Dst: CheckPoint50_6c:96:3f (08:00:27:fc:96:3f)
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 151.139.128.14
- Transmission Control Protocol, Src Port: 64879, Dst Port: 80, Seq: 0, Len: 0

Answer: 00:1c:7f:6c:96:3f

E11. Examine packet 416 in the network file and identify the protocol being used.

traffic.pcapng

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No.	Time	Source	Destination	Protocol	Length	Info
407	20.409663	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
408	20.409683	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
409	20.409894	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
410	20.409963	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
411	20.500090	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
412	20.500177	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
413	20.504026	10.103.50.116	10.103.255.255	SHL_NE...	280	SAW LOGON request from Client[Malformed Packet]
414	20.508046	204.79.197.200	10.103.51.159	TLSv1.2	198	Application Data
415	20.508109	10.103.51.159	10.103.51.159	TCP	54	59023 → 443 [ACK] Seq=26828 Ack=5380 Min=261888 Len=0
416	20.508312	10.103.51.159	10.103.0.20	DNS	86	Standard query 807433 A checkapplecare.microsoft.com
417	20.63261	10.103.51.159	10.103.0.20	DNS	86	Standard query 807433 A checkapplecare.microsoft.com
418	20.651419	10.103.0.20	10.103.51.159	DNS	209	Standard query response 807433 A checkapplecare.microsoft.com CNAME wd-prod-ss-trafficmanager.net CNAME wd-prod-ss-eu-west-...
419	20.658534	10.103.51.159	10.103.51.159	TCP	60	59023 → 443 [RST] Seq=5380 Win=0 Len=0
420	20.705932	10.103.51.159	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
421	20.726436	10.103.51.159	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
422	20.726538	10.103.51.159	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
423	20.726538	10.103.51.159	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
424	20.726538	10.103.51.159	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=1903 Min=4203264 Len=0
425	20.775040	10.103.50.26	10.103.255.255	NBNS	92	Name query NB 507H-021C>
426	20.775987	10.103.50.25	10.103.255.255	NBNS	92	Name query NB 507H-021C>

Frame 416: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface DeviceNPF {53...} KX J E

Ethernet II, Src: Hewlett-Packard 78:cd:da (10:60:4b:78:cd:da), Dst: Hewlett-Packard 08:07:29 (00:15:60:08:07:29)

Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20

User Datagram Protocol, Src Port: 59023, Dst Port: 53

Domain Name System (query)

Profile Default 6:26 PM 7/25/2016

Answer: DNS

E12. Examine packet 416 in the network file and identify the destination IP address.

The image shows a Wireshark network traffic capture. The main pane displays a list of packets, with packet 416 selected. The packet details pane shows the structure of the packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and Domain Name System (query) header. The destination IP address is 10.103.51.159.

No.	Time	Source	Destination	Protocol	Length	Info
407	20.499463	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
408	20.499463	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
409	20.499463	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
410	20.499463	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
411	20.499463	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
412	20.500099	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
413	20.500177	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
414	20.504026	10.103.50.116	10.103.255.255	SNB_NE...	280	SAW LOGON request from client[malformed packet]
415	20.508046	204.79.197.200	10.103.51.159	TLSv1.2	198	Application Data
416	20.508046	204.79.197.200	10.103.51.159	TLSv1.2	54	59023 → 443 [ACK] Seq=5236 Ack=19363 Win=4203264 Len=0
417	20.508046	204.79.197.200	10.103.51.159	DNS	86	Standard query 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
418	20.508046	204.79.197.200	10.103.51.159	DNS	289	Standard query response 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
419	20.508046	204.79.197.200	10.103.51.159	DNS	86	Standard query 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
420	20.508046	204.79.197.200	10.103.51.159	DNS	289	Standard query response 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
421	20.508046	204.79.197.200	10.103.51.159	DNS	86	Standard query 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
422	20.508046	204.79.197.200	10.103.51.159	DNS	289	Standard query response 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
423	20.508046	204.79.197.200	10.103.51.159	DNS	86	Standard query 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
424	20.508046	204.79.197.200	10.103.51.159	DNS	289	Standard query response 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
425	20.508046	204.79.197.200	10.103.51.159	DNS	86	Standard query 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...
426	20.508046	204.79.197.200	10.103.51.159	DNS	289	Standard query response 0x7433 A checkappsec.microsoft.com CNAME wd-prod-ss.trafficmanager.net CNAME wd-prod-ss-eu-west-...

Answer: 10.103.0.20

E13. Examine the network file and analyze the first HTTP packet; what is the source IP address?

traffic-pcapng

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HTTP

No.	Time	Source	Destination	Protocol	Length	Info
19807	228.266244	10.103.51.159	151.139.128.14	HTTP	290	GET /HEWITTZBMREWS/TAJBGU/PDGCgUAB88aht08FzTVa8aWvzJ38QUQHY9aQU659X2BAJ36pV8HtG87J1V7MkYHtQCECS8N8oK8p55ZFpJ9HtKfN8K8K...
19810	228.266245	151.139.128.14	10.103.51.159	OKSP	781	Response
19812	228.332276	10.103.51.159	151.139.128.14	HTTP	292	GET /HEWITTZBMREWS/TAJBGU/PDGCgUAB855ax66447YmK8HPVU1UL3vK8ZQQKZFgZ74pKZ8Uv5pmq4Z6Z7me712h1CCMaGQu637XaH1LEbEIdD...
19817	228.389784	151.139.128.14	10.103.51.159	OKSP	563	Response
19802	269.746721	10.103.51.159	23.221.154.40	HTTP	271	GET /singleTitle/summary/alias/experiencebyname/today?markket=en-us&source=approximate&tenant=amp&vertical=sports HTTP/1.1
19804	269.749357	23.221.154.40	10.103.51.159	HTTP	315	HTTP/1.1 301 Moved Permanently
20767	305.694964	10.103.51.159	10.103.51.159	HTTP	1169	GET / HTTP/1.1
20885	304.618195	10.103.51.159	10.103.51.159	HTTP	1228	GET /img/PCtch8u8798e083x7089F6765578d6cf.png HTTP/1.1
20886	304.618196	10.103.51.159	10.103.51.159	HTTP	1228	GET /img/PCtch8u8798e083x7089F6765578d6cf.png HTTP/1.1
20891	304.821164	10.103.51.159	10.103.51.159	HTTP	1409	HTTP/1.1 200 OK (text/html)
20892	304.821168	10.103.51.159	10.103.51.159	HTTP	1216	GET /img/flexible/logo/pc/result82.png HTTP/1.1
20919	305.183955	10.103.51.159	10.103.51.159	HTTP	1219	GET /img/flexible/logo/pc/naak-result.png HTTP/1.1
20937	305.186294	10.103.51.159	10.103.51.159	HTTP	1493	HTTP/1.1 200 OK (PNG)
20951	305.187355	10.103.51.159	10.103.51.159	HTTP	389	HTTP/1.1 200 OK (PNG)
20960	305.190240	10.103.51.159	10.103.51.159	HTTP	1163	HTTP/1.1 200 OK (PNG)
21007	305.544293	10.103.51.159	10.103.51.159	HTTP	1053	HTTP/1.1 200 OK (PNG)
21040	305.694886	10.103.51.159	110.242.69.147	HTTP	467	GET /static/superman/img/qrcode/qrcode&2-daf987ad92.png HTTP/1.1
21043	305.710660	10.103.51.159	110.242.69.147	HTTP	469	GET /static/superman/img/topnav/saladogun&2-ed8979e0e.png HTTP/1.1
21046	305.719284	10.103.51.159	110.242.69.147	HTTP	469	GET /static/superman/img/topnav/zhidogun&2-ed842f6c4.png HTTP/1.1
21045	305.722196	10.103.51.159	110.242.69.147	HTTP	467	GET /static/superman/img/topnav/zhidogun&2-ed842f6c4.png HTTP/1.1

> Frame 19807: Packet, 290 bytes on wire (2320 bits), 290 bytes captured (2320 bits) on interface DeviceM

> Ethernet II, Src: Hewlett-Packa_78icd4a (10:60:4b:78:icd4a), Dst: Checkpoint50_6c:96:3f (00:1c:7f:6c:96:3f)

> Internet Protocol Version 4, Src: 10.103.51.159, Dst: 151.139.128.14

> Transmission Control Protocol, Src Port: 64079, Dst Port: 80, Seq: 1, Ack: 1, Len: 286

> Hypertext Transfer Protocol

Profile: Default

Packets: 26455 - Displayed: 238 (0.9%)

ENG IN 6:28 PM 7/25/2026

Answer: 10.103.51.159

E14. Examine the network file and analyze the first HTTP packet; what is the source port?

The image shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows a single packet (No. 19807) of type HTTP, source 10.103.51.159, destination 151.139.128.14, length 290. The packet details pane on the right shows the following structure:

- Frame 19807: Packet, 290 bytes on wire (2320 bits), 290 bytes captured (2320 bits) on interface DeviceM
- Ethernet II, Src: Hewlett-Packard_78:cd:4a (10:60:4b:78:cd:4a), Dst: Checkpoint_6c:96:3f (08:1c:7f:6c:96:3f)
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 151.139.128.14
- Transmission Control Protocol, Src Port: 64079, Dst Port: 80, Seq: 1, Ack: 1, Len: 286
- Hypertext Transfer Protocol

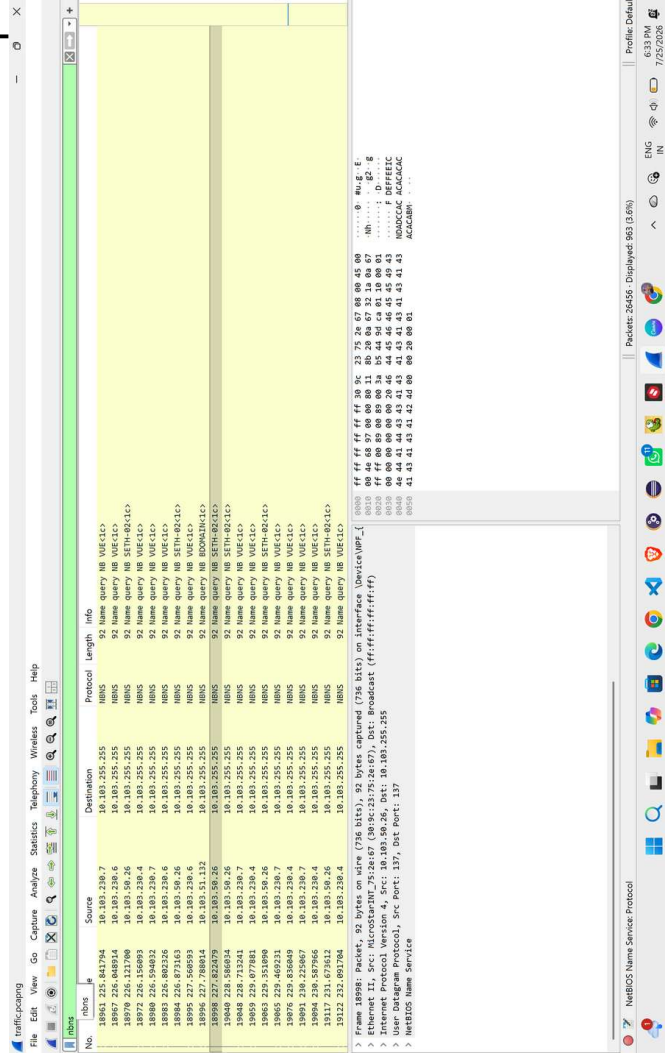
The packet bytes pane shows the raw data of the HTTP request, including the GET request line and various headers.

Answer: 10.103.51.159

E15. Analyze the network file and find the duration of the capture (in seconds).

Ans: 347.58 seconds

E16. Analyze the network file and find the number of NBNS packets.



Answer: 963 packets

E17. Analyze the network file and find the number of TCP packets sent with source port 443.

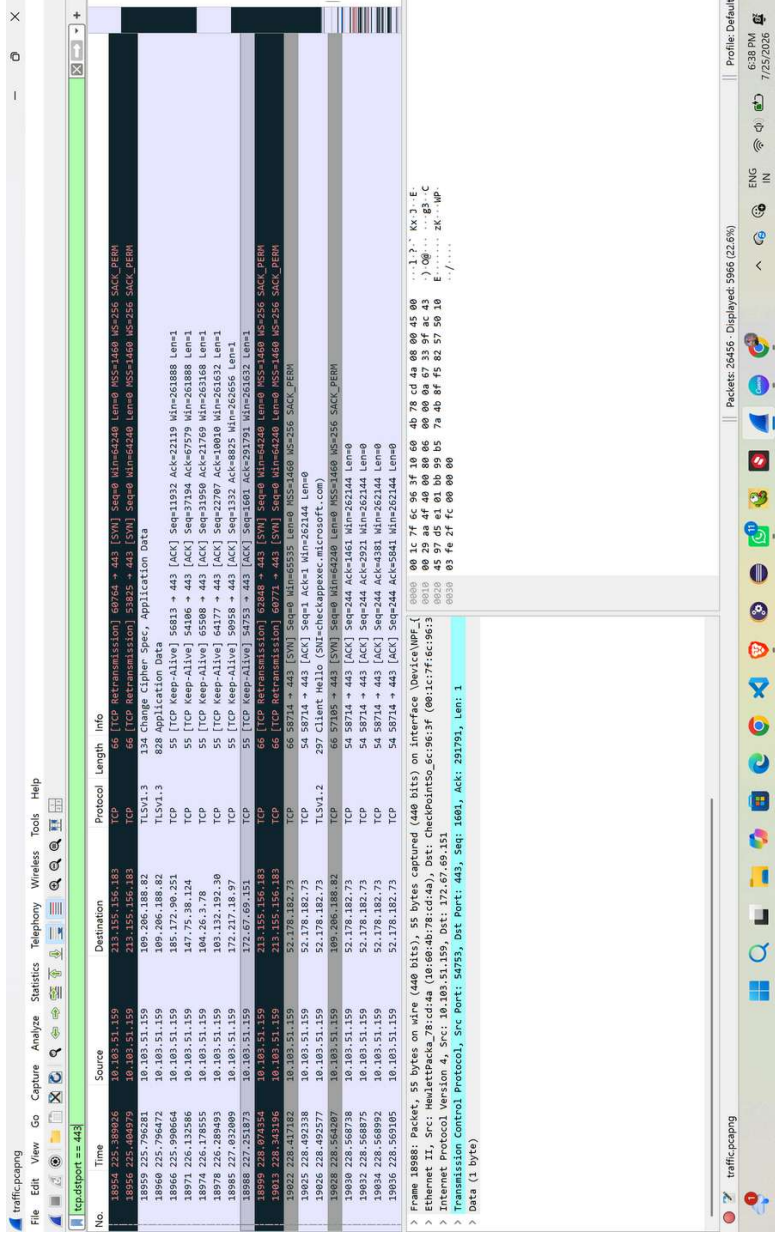
The image shows a Wireshark packet capture analysis. The top pane displays a list of packets. Packet 19037 is highlighted in yellow, indicating it is the selected packet. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
18952	225.238955	142.250.185.238	10.103.51.159	TCP	60	443 → 59481 [ACK] Seq=7841 Ack=6925 Win=82176 Len=0
18958	225.795847	189.206.188.82	10.103.51.159	TLSv1.3	310	Server Hello, Change Cipher Spec, Application Data
18962	225.698246	189.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=728 Win=65535 Len=0
18963	225.698432	189.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=728 Win=65535 Len=0
18968	226.691985	185.172.90.251	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 56813 [ACK] Seq=22119 Ack=11933 Win=64128 Len=0 SLE=11932 SRE=11933
18976	226.237097	184.26.3.78	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 65808 [ACK] Seq=21769 Ack=31951 Win=139264 Len=0 SLE=31950 SRE=31951
18977	226.263888	147.75.38.124	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54288 [ACK] Seq=27579 Ack=37195 Win=35704 Len=0 SLE=37194 SRE=37195
18981	226.698412	185.132.182.36	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 64459 [ACK] Seq=8808 Ack=22788 Win=16164 Len=0 SLE=1323 SRE=1323
18982	226.698412	185.132.182.36	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 64459 [ACK] Seq=8808 Ack=22788 Win=16164 Len=0 SLE=1323 SRE=1323
18989	227.508460	172.67.69.151	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54753 [ACK] Seq=20191 Ack=1602 Win=50632 Len=0 SLE=1601 SRE=1602
19024	228.628669	52.178.182.73	10.103.51.159	TCP	66	443 → 57104 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SCS=556 SACK_PERM
19027	228.593828	189.206.188.82	10.103.51.159	TCP	60	443 → 51904 [RST, ACK] Seq=1 Ack=518 Win=65535 Len=0
19029	228.568880	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1 Ack=244 Win=525312 Len=1460 [TCP RST reassembled in 19037]
19031	228.568851	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1661 Ack=244 Win=525312 Len=1460 [TCP RST reassembled in 19037]
19033	228.568864	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=2921 Ack=244 Win=525312 Len=1460 [TCP RST reassembled in 19037]
19035	228.569076	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=4381 Ack=244 Win=525312 Len=1460 [TCP RST reassembled in 19037]
19037	228.569186	52.178.182.73	10.103.51.159	TLSv1.2	1291	Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
19041	228.628877	189.206.188.82	10.103.51.159	TCP	60	443 → 57105 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SCS=556 SACK_PERM
19044	228.651282	52.178.182.73	10.103.51.159	TLSv1.2	185	Change Cipher Spec, Encrypted Handshake Message
19045	228.726071	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7125 Ack=2326 Win=525568 Len=0

The bottom pane shows the details of the selected packet (19037). The Ethernet II section shows the source MAC address as 08:00:27:64:8c:3f and the destination MAC address as 08:00:27:64:8c:3f. The Internet Protocol section shows the source IP as 172.67.69.151 and the destination IP as 10.103.51.159. The Transmission Control Protocol section shows the source port as 443 and the destination port as 54753. The packet is 1602 bytes long.

Answer: 7275 packet

E18. Examine the network file and find the number of TCP packets sent to destination port 443



Answer: 59666 packets

E19. Examine the network file; what is the number of packets sent to the destination IP address 204.79.197.200?

The image shows a Wireshark packet capture of a network session. The top pane displays a list of 116 packets, all of which are sent to the destination IP address 204.79.197.200. The middle pane shows the details of the selected packet (packet 116), which is a TCP segment. The bottom pane shows the raw packet data in hexadecimal and ASCII. The status bar at the bottom indicates that 116 packets are displayed.

No.	Time	Source	Destination	Protocol	Length	Info
15766	202.621872	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=56996 Ack=5128 Win=262144 Len=0
15767	202.624838	10.103.51.159	204.79.197.200	TLSv1.2	1598	Application Data
15772	202.742413	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=58535 Ack=6588 Win=262144 Len=0
15776	203.336641	10.103.51.159	204.79.197.200	TLSv1.2	247	Application Data
15783	203.492651	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=58728 Ack=7812 Win=268864 Len=0
15784	203.516462	10.103.51.159	204.79.197.200	TLSv1.2	247	Application Data
15785	203.607558	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=59116 Ack=8998 Win=262144 Len=0
15792	203.690724	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=59310 Ack=10141 Win=268864 Len=0
15793	203.784620	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10225 Win=268864 Len=0
15800	207.593527	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10225 Win=268864 Len=0
15802	207.679645	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10225 Win=268864 Len=0
15841	207.586205	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
15842	207.586278	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
15843	207.586351	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
15844	207.586399	10.103.51.159	204.79.197.200	TLSv1.2	238	Application Data
15846	207.588838	10.103.51.159	204.79.197.200	TLSv1.2	538	Application Data
15847	207.588898	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
15861	207.588422	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10183 Win=268864 Len=0
15862	207.593527	10.103.51.159	204.79.197.200	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10225 Win=268864 Len=0

Summary of packets sent to 204.79.197.200:

- Frame 15982: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface DeviceNPF...
- Ethernet II, Src: Hewlett-Packard_78:cd:4a (10:60:4b:78:cd:4a), Dst: Checkpoint_6c:96:3f (00:1c:7f:6c:96:3f)
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 204.79.197.200
- Transmission Control Protocol, Src Port: 51528, Dst Port: 443, Seq: 125891, Ack: 10183, Len: 0

Answer: 116 packets

E20. Analyze the network file and find the MAC address associated with the IP address 204.79.197.200. To which vendor does this MAC address belong?

The image shows a Wireshark packet capture analysis. The packet list pane displays a series of TCP packets from source 204.79.197.200 to destination 10.103.51.159. Packet 25276 is selected, showing a SYN packet with sequence 352,997,685 and window 0. The packet details pane shows the Ethernet II frame and Internet Protocol Version 4 fields. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
15884	207.592049	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=108804 Win=4284800 Len=0
15885	207.592192	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=110244 Win=4284800 Len=0
15886	207.592192	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=111684 Win=4284800 Len=0
15887	207.592398	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=113124 Win=4284800 Len=0
15888	207.592425	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=114564 Win=4284800 Len=0
15889	207.592489	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=116004 Win=4284800 Len=0
15890	207.592619	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=117444 Win=4284800 Len=0
15891	207.592709	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=118884 Win=4284800 Len=0
15892	207.592769	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=120324 Win=4284800 Len=0
15893	207.592857	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=121764 Win=4284800 Len=0
15894	207.593071	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=123204 Win=4284800 Len=0
15895	207.593205	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=124644 Win=4284800 Len=0
15896	207.593253	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=126084 Win=4284800 Len=0
15897	207.593302	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=127524 Win=4284800 Len=0
15898	207.593362	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=128964 Win=4284800 Len=0
15899	207.593505	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [ACK] Seq=10183 Ack=130404 Win=4284800 Len=0
15900	207.593527	10.103.51.159	204.79.197.200	TLSv1.2	96	Application Data
15901	207.593575	204.79.197.200	10.103.51.159	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10225 Win=268864 Len=0
15902	207.593643	10.103.51.159	204.79.197.200	TLSv1.2	198	Application Data
15903	207.593705	204.79.197.200	10.103.51.159	TCP	54	51528 → 443 [ACK] Seq=125891 Ack=10369 Win=268864 Len=0
25276	352.997685	204.79.197.200	10.103.51.159	TCP	60	443 → 51528 [SYN, ACK] Seq=10183 Ack=125931 Win=0 Len=0

Packet 25276 details:

- Ethernet II, Src: Hewlett-Packard_78:cd:da (10:60:4b:78:cd:da), Dst: Checkpoint_0c:96:3f (00:1c:7f:6c:96:3f)
- Internet Protocol Version 4, Src: 204.79.197.200, Dst: 10.103.51.159, Len: 60
- Transmission Control Protocol, Src Port: 51528, Dst Port: 443, Seq: 10183, Ack: 10183, Len: 0

Answer: MAC: 00:1c:7f:6c:96:3f